

Energy Consumption Predictor Project Plan

1. Project Overview

Build an IoT-based system to simulate building energy usage, store data, train an ML model, and display predictions on a website.

2. Modules

A. IoT Simulation (Wokwi)

- Simulate rooms and devices (AC, fan, lights)
- Generate timestamped power usage data

B. Data Collection

- Send data via HTTP to Flask server
- Store data in CSV format

C. Data Processing

- Convert raw data to daily/monthly consumption
- Feature engineering (kWh calculation, aggregation)

D. Machine Learning

- Train regression model (Linear/Random Forest)
- Predict monthly energy consumption

E. Backend (Flask)

- API endpoints: /data, /predict

F. Frontend (Next.js + Tailwind)

- Dashboard with graphs
- Prediction display

3. Timeline

Week 1: Simulation

Week 2: Data pipeline

Week 3: ML model

Week 4: Backend

Week 5: Frontend

4. Deliverables

- CSV dataset
- Trained ML model
- API backend
- Web dashboard

5. Bonus Features

- Cost optimization suggestions
- Peak usage alerts